

Class XII | Unit 6 | weight 15%

Introduction to multimedia

Board weight about 15% | 20 theory periods and 20 lab periods

Multimedia is the use of more than one medium, such as text, picture, sound and moving image, in one presentation. This lecture names the parts, the file types, the software, and the steps that produce a short video.

What you should be able to do

- Define multimedia and explain its importance.
- Describe text, graphics, animation, audio and video, and name file types for each.
- Distinguish web-based and non-web multimedia applications, and name editing tools.
- Use the ideas of scene, layer and keyframe, and describe export and sharing.

Path through the lecture

- 1 Play a one-minute clip and ask students to name every medium they can see or hear.
- 2 Attach a file extension to each medium before opening the editor.
- 3 Build a 20-second version in the lab first. The curriculum's two-minute video is the full project.

6.1 What multimedia is

Multimedia. The combination of two or more media, such as text, graphics, animation, audio and video, in a single digital presentation.

A textbook page of plain text is one medium. A lesson that shows a labelled diagram, speaks the definition and plays a short clip is multimedia. Colleges use it for lessons, laboratory briefings and presentations. It helps a learner who follows a picture more easily than a paragraph, and it lets one package carry the explanation, the example and the sound.

- A difficult process can be shown as well as described.
- The same lesson can be watched again.
- Text can be searched, while the picture carries the part that is hard to search.
- A presentation can serve a class in the room and a class joining over a network.

6.2 Components and file types

The five components

Component	What it contributes	File types to remember
Text	Titles, labels and definitions	txt, doc, pdf
Graphics	Still pictures and diagrams	jpg, png, gif
Animation	A sequence of drawn or generated frames	gif, and video formats that hold the result
Audio	Speech, music and effects	mp3, wav
Video	Moving pictures, usually with sound	mp4, avi

GIF can be a still picture or a short animation. PNG keeps sharp diagrams and can include transparency. JPEG suits photographs and is lossy: it saves space by discarding some detail. WAV is a full recorded waveform and is large. MP3 is compressed audio. MP4 is the usual container for a finished student video. Use these names in answers even if the lab also produces other types.

6.3 Applications and packages

A **web-based** multimedia application runs in a browser or is delivered as a site: an online lesson, a video page, an interactive diagram. A **non-web** application runs as a program on the computer: a slide show, a video editor, a player. The curriculum names packages such as ProShow Gold for photo and slide video, and JetAudio for playing and working with audio. Current labs often use OpenShot or Shotcut for video and Audacity for sound. Name the curriculum examples, and also name the tool you actually used in the practical.

6.4 Making the video

Scene or canvas. The frame in which you place the parts of the presentation. It is the visible area the viewer will see.

Layer. One level of content stacked with others. A background image can sit on a lower layer and a title on a higher layer, so the title remains editable on its own.

Keyframe. A frame in which you set a property, such as position or size. The software creates the in-between frames, which produces the motion.

- 1 Open a new project and set the canvas to a standard frame such as 1280 by 720.
- 2 Import or record the pictures, the voice and any existing clip. Keep the source files in one project folder.
- 3 Place the background on a lower layer and titles and labels on higher layers.
- 4 Set keyframes where an object should start and where it should finish. Keep motion slow enough to read.
- 5 Record or import narration. Match the length of each sentence to the picture it describes.
- 6 Watch the whole piece from the start and repair a wrong order before you export.
- 7 Export to MP4 for sharing, and keep a second export such as AVI if the question asks for more than one format.
- 8 Upload the finished file to the sharing place your college uses, or to a video site the teacher names, and send the link to the class. Respect the privacy of any person who appears in the clip, and use only pictures and music you have the right to use.

The curriculum project is a video of about two minutes on a topic from this course, for example the OSI layers or the states of a process. Two minutes is long enough to need a plan and short enough to finish in the lab periods.

In the lab

Plan the two minutes on paper first: four scenes of about 30 seconds. Scene 1 states the definition. Scene 2 labels a diagram. Scene 3 gives an example. Scene 4 repeats the definition in one sentence. Then open the editor.

Key terms

Term	Meaning to write in an answer
Multimedia	More than one medium combined in one presentation.
Graphics	Still images.
Animation	Change across frames that creates the sense of motion.
Layer	A separate level of content stacked in a scene.
Keyframe	A frame where a property is set so the software can fill the frames between.
Export	Saving the finished project as a playable file such as MP4.

Bring it together

- Multimedia combines text, graphics, animation, audio and video.
- Learn one file type for each component.
- Web applications are delivered in the browser. Desktop editors such as a video or audio package are non-web applications.
- A scene holds layers. Keyframes set motion. Export, then share with permission to use the material.

Do this in the practical

- 1 Place a title and an imported image on separate layers and save the picture in two formats.
- 2 Import one audio file and one image into one project.
- 3 Add two keyframes that move a label across the canvas.
- 4 Produce a short video, export it as MP4, and show the file to your teacher. Build toward the two-minute piece.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. Multimedia means:

- A. Using more than one medium together
- B. Using only plain text
- C. A type of RAM
- D. A bus topology

2. Which extension is typical of a photograph?

- A. jpg
- B. mp3
- C. txt only
- D. sql

3. A keyframe is:

- A. A frame where you set a property for the animation
- B. A primary key
- C. A keyboard port
- D. An operating system

4. A layer is used so that:

- A. Parts of a scene can be edited separately
- B. RAM is refreshed
- C. A file is binary
- D. A pointer is declared

5. MP3 is:

- A. Compressed audio
- B. A still photograph standard
- C. A C++ keyword
- D. A normal form

6. A web-based multimedia application:

- A. Is delivered through a browser or the web
- B. Cannot contain sound
- C. Is always stored in ROM
- D. Replaces the DBMS

7. Exporting a video means:

- A. Saving the finished presentation as a playable file
- B. Deleting the project
- C. Printing the motherboard
- D. Normalising a table

8. PNG is most suitable in class for:

- A. A sharp diagram
- B. A database key
- C. A process state only
- D. A twisted-pair cable

Answer key

Q	Answer	Why this is the answer
1	A	The word names a combination of media.
2	A	JPEG is a common photographic image format.
3	A	The editor interpolates between keyframes.
4	A	Stacked layers keep the title independent of the background.
5	A	MP3 is an audio format.
6	A	Web-based means the user reaches it through the web.
7	A	Export produces the file you can play or upload.
8	A	PNG keeps crisp edges, which suits diagrams and text in pictures.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Define multimedia and give two advantages.**Model answer**

Multimedia combines two or more media, such as text, sound and video, in one presentation. It can show a process that is hard to follow from text alone, and the learner can play the lesson again. A labelled diagram with a spoken definition is a simple example.

Q2. Name the five components and one file type for each.**Model answer**

Text uses txt or pdf. Graphics use jpg or png. Animation may use gif. Audio uses mp3 or wav. Video uses mp4 or avi. The component is the kind of content. The file type is how that content is stored.

Q3. Differentiate a web-based and a non-web multimedia application.

Model answer

A web-based application is reached through a browser, such as an online lesson with video. A non-web application is a program on the computer, such as a video editor or a desktop player. Both can contain the same five components. They differ in how the user starts them.

Q4. Define scene, layer and keyframe.

Model answer

The scene or canvas is the visible frame. A layer is one stacked level of content, so a title can sit above a picture. A keyframe is a frame where a property such as position is set. The software fills in the frames between keyframes to make movement.

Q5. Why keep source files in one folder before exporting?

Model answer

The project refers to the imported pictures and sounds by their location. If a file is moved, the editor cannot find it and the export is incomplete. One project folder keeps the path stable until the video has been exported.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain the components of multimedia and the file types used to store them.

- 1 Define multimedia and give its importance in one short paragraph.
- 2 Explain text, graphics, animation, audio and video, each with a classroom example.
- 3 Attach at least one file type to each, and say when jpg is a better photo format than png is for a diagram, or keep the comparison inside the graphics row.
- 4 Distinguish a web delivery and a desktop package, naming one package.

Question 2. Describe how you would make a two-minute educational video and share it.

- 1 Plan four short scenes on a syllabus topic.
- 2 Set the canvas, import media, and arrange layers.
- 3 Use keyframes for one simple movement and add narration.
- 4 Export to MP4 and name a second format.
- 5 Upload or share the file with the teacher's method, and use only material you have permission to use.